Math 2001 Analysis Exercises

If $f : \mathbb{R} \to \mathbb{R}$ is a function that can be integrated (such as $e^x, \sin x, 1 - x^2$, etc) we let $\int_S f(x)\,dx$ denote the integral of $f(x)$ over the set S. For example, $\int_{[0,1]} f(x)\,dx = \int_0^1 f(x)\,dx$ and

$$\int_{[0,1/4]\cup[3/4,1]} f(x)\,dx = \int_0^{1/4} f(x)\,dx + \int_{3/4}^1 f(x)\,dx.$$

1. Give an example of a nonzero function $f(x)$ and a closed set S to such that

$$\int_{[0,1]} f(x)\,dx = \int_{[0,1]\setminus S} f(x)\,dx$$

where $[0,1] \setminus S$ denotes the set $[0,1]$ without S. Include some indication for why your example works!

2. Give a second example of a nonzero function $f(x)$ and a nonempty open set $\mathcal{O}$ is contained in $[0,1]$ such that

$$\int_{[0,1]} f(x)\,dx = \int_{[0,1]\setminus \mathcal{O}} f(x)\,dx.$$

Include some indication for why your example works!

3. Take an educated guess of the value of $\int_{[0,1]\cap\mathbb{Q}} x^2\,dx$ where $[0,1] \cap \mathbb{Q}$ is the set of real numbers between 0 and 1 that are not rational (not expressible as a/b for positive integers a, b with $a < b$).

4. Take an educated guess of the value of $\int_C x^2\,dx$ where C is the Cantor set.